DGMS 2019 • SECTION G: SEISMOLOGY & EARTH INTERIOR

Question 64: Seismic Wave Velocity Distribution Within the Earth (PREM Model)

Complete Velocity Range from Upper Crust to Deep Mantle, Elastic Moduli, and PREM Radial Profiles

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OFFICIAL EXAMINATION QUESTION

The velocity of the seismic waves within the earth ranges from

(a) 1 to 4 km/second

(b) 2 to 8 km/second

(c) 3 to 10 km/second

(d) **2 to 13 km/second [CORRECT]**

VERIFIED ANSWER KEY

Correct Option: (d) 2 to 13 km/second — Throughout the Earth's interior (from the near-surface crustal layers to the base of the lower mantle), seismic wave velocities span from **~2.0–3.0 km/s** (shear waves in shallow crust and sedimentary cover) up to **~13.6–13.7 km/s** (maximum compressional P-wave velocity at the base of the lower mantle / D'' layer, just above the core-mantle boundary).

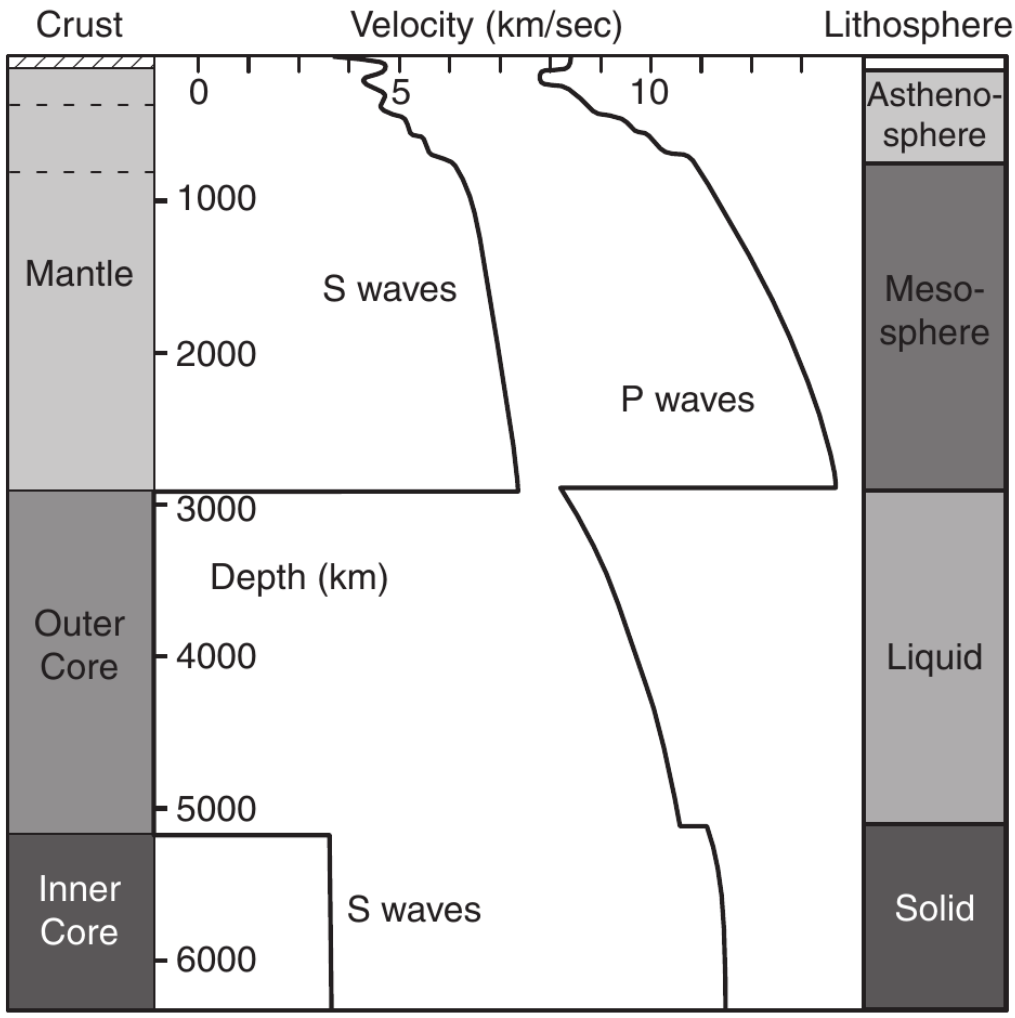


FIGURE 3 Variation in *P* and *S* wave velocities with depth. Compositional subdivisions of the Earth are on the left, and rheological subdivisions are on the right. After Kearey and Vine (1990). Reprinted by permission of Blackwell Science, Inc.

Figure 64.1 | Radial Seismic Velocity Profile of the Entire Earth: Compressional (P) and shear (S) wave velocities plotted as a function of depth from the surface (0 km) to the Earth's center (6370 km). In the crust, $V_s \approx 2\text{--}3.5\text{ km/s}$ and $V_p \approx 5\text{--}6.5\text{ km/s}$. Across the mantle, V_p increases progressively from 8.0 km/s at the Moho to a global terrestrial peak of **13.7 km/s** at the base of the lower mantle (~2898 km). (Source: John D. Winter, *Principles of Igneous and Metamorphic Petrology*, 2nd Ed., Fig. 1-3, p. 7; after Kearey & Vine, 1990; PREM model).

1. Elastic Wave Physics & Seismic Velocity Controls

The velocities of body seismic waves propagating through isotropic solid elastic media depend directly on the density (ρ) and elastic moduli: bulk modulus (K, incompressibility) and shear modulus (μ, rigidity):

- **Compressional / Primary Wave Velocity (V_p):** $$V_p = \sqrt{\frac{K + \frac{4}{3}\mu}{\rho}}$$
- **Transverse / Shear Wave Velocity (V_s):** $$V_s = \sqrt{\frac{\mu}{\rho}}$$

Although density ρ increases downward with depth, the elastic moduli K and μ increase much more rapidly due to intense self-gravitational compaction and pressure. Consequently, both V_p and V_s increase with depth throughout the crust and mantle.

2. Detailed Layer-by-Layer Velocity Breakdown (PREM Model)

Geological Layer	Depth Range (km)	V_s Velocity Range (km/s)	V_p Velocity Range (km/s)	Physical & Mineralogical Characteristics
Upper Crust & Sediments	0 – 15 km	**1.5 – 3.2 km/s**	**2.5 – 6.0 km/s**	Porous sediments, fractured granites/basalts; lowest velocities in Earth.
Lower Crust	15 – 35 km	3.4 – 3.9 km/s	6.5 – 7.1 km/s	High-grade granulites and amphibolites.
Uppermost Mantle (Moho)	35 – 220 km	4.4 – 4.6 km/s	**8.0 – 8.2 km/s**	Peridotite (olivine + pyroxene). Moho jump from 7 to 8 km/s.
Transition Zone	410 – 660 km	4.8 – 5.5 km/s	8.8 – 10.2 km/s	Phase transitions: Olivine $\rightarrow$ Wadsleyite (410) $\rightarrow$ Ringwoodite (520).
Lower Mantle (Mesosphere)	660 – 2700 km	5.5 – 7.2 km/s	10.7 – 13.5 km/s	Bridgmanite + ferropericlase; uniform compaction increase.

Geological Layer	Depth Range (km)	V_s Velocity Range (km/s)	V_p Velocity Range (km/s)	Physical & Mineralogical Characteristics
Core-Mantle Boundary ($D"$)	2700 – 2898 km	7.2 – 7.3 km/s	**13.6 – 13.7 km/s**	**Global maximum V_p velocity in Earth!** Heterogeneous thermal boundary layer.
Outer Core	2898 – 5145 km	**0.0 km/s** ($\mu = 0$)	8.1 – 10.3 km/s	Molten liquid iron alloy; P-wave drops abruptly from 13.7 to 8.1 km/s.
Inner Core	5145 – 6370 km	3.5 – 3.7 km/s	11.0 – 11.3 km/s	Solid crystalline iron-nickel alloy; S-waves reappear.

3. Comprehensive Option Analysis

- **(a) 1 to 4 km/second:** *Incorrect.* This range corresponds only to surface waves (Rayleigh/Love) or shear waves in shallow unconsolidated sediment. It completely overlooks mantle and core velocities ($>8\text{ km/s}$).
- **(b) 2 to 8 km/second:** *Incorrect.* This represents only the crust and uppermost mantle just beneath the Moho ($\sim 8.1\text{ km/s}$). It omits the entire transition zone and lower mantle where velocities reach nearly 14 km/s.
- **(c) 3 to 10 km/second:** *Incorrect.* It omits shallow slow waves ($<3\text{ km/s}$) as well as the deep mantle velocities ($10\text{ to }13.7\text{ km/s}$).
- **(d) 2 to 13 km/second: Correct.** The universally recognized whole-Earth body wave velocity range spans from ~2 km/s (near-surface crustal shear waves) up to ~13–13.7 km/s (deep lower mantle compressional waves).

4. Authoritative Textbook References

- **John D. Winter (2014)** *Principles of Igneous and Metamorphic Petrology*, 2nd Ed., Pearson, Chapter 1 ("Some Fundamental Concepts"), Section 1.2 ("The Internal Structure of the Earth"), pp. 6–8, Fig. 1-3.
- **Dziewonski & Anderson (1981)** "Preliminary Reference Earth Model (PREM)", *Physics of the Earth and Planetary Interiors*, 25(4): 297–356.

- **P. Kearey, K. A. Klepeis & F. J. Vine (2009)** *Global Tectonics*, 3rd Ed., Wiley-Blackwell, Chapter 2 ("The Interior of the Earth"), pp. 19–35.
- **Peter M. Shearer (2009)** *Introduction to Seismology*, 2nd Ed., Cambridge University Press, Chapter 3 ("Seismic Waves"), pp. 38–62.